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Department of Electrical Engineering		
National Institute of Technology Kurukshetra		
Course : AC Machine	Semester: 3rd	Total Questions : 8
Course Code : EEPC201	Date: 2.12.25	Question to attempt: 5
Program: BTech (EE)	Time: 9.30 am	Max. Marks: 50
Exam : End Semester	Centre : E101,E102	Max. Time : 3 hours

Note: Assume the necessary data if required

Q. No	Question	Marks
1)	Explain the working principle of a single-phase transformer. Derive the equation of induced emf on primary and secondary side. Draw the phasor diagram of a single-phase transformer at leading power factor.	10
2) (a)	What is the voltage regulation of 1- ϕ transformer? Derive the condition for maximum and minimum voltage regulation in per unit system.	5
(b)	A 50kVA, 2200/110V transformer has the following test results: OC test measurement, LV side: 400W, 10A, 110V SC test measurement on HV side: 808W, 20.5A, 90V Compute the parameter of the equivalent circuit referred to HV side. Also calculate the % Voltage regulation and efficiency at full load and 0.8pf lagging.	5
3) (a)	Explain the construction and working principle of an autotransformer with neat diagram. Also explain its advantages and disadvantages over two-winding transformer.	5
(b)	Explain the parallel operation of transformers with neat diagram and state the necessary and desirable conditions for parallel operation. What will be effect on parallel operation if desirable conditions are not fulfilled.	5
4) (a)	Explain the running light and blocked rotor test of a three-phase induction motor.	5
(b)	A 10kW, 400V, 4 pole delta-connected squirrel cage induction motor gave the following test result: No-load test: 400V, 8A, 250 watts. Blocked- rotor test: 90V, 35A, 1350 watts. The DC resistance of the stator winding per phase measured immediately after the blocked rotor test is 0.6 ohms. Calculate the rotational losses and the equivalent circuit parameters.	5
5) (a)	Explain various methods of starting of 3-ϕ induction motor. Also compare the starting current and torque of each method w.r.t direct on line starting.	5
(b)	A small squirrel-cage induction motor has a starting current of six times the full-load current and a full-load slip of 0.05. Find in pu of full-load values, the current (line) and starting torque with the following methods of starting (a) direct switching. (b) stator-resistance starting with motor current limited to 2 pu, (c) autotransformer starting with motor current limited to 2 pu, and (d) star-delta starting.	5

6) (a)	Explain the various method of speed control of 3-ϕ induction motor.	5
(b)	A 440 V, 50 Hz, 4-pole 3-phase, delta-connected motor has a leakage impedance of (0.3 + j 5.5 + 0.25/s) ohm/phase (delta phase) referred to the stator. The stator to rotor voltage ratio is 2.5. Determine the external resistance to be inserted in each star phase of the rotor winding such that the motor develops a gross torque of 150 Nm at a speed of 1250 rpm.	5
7)	From the equivalent circuit of a poly-phase induction motor, obtain the following relations:	10
	(a) $\frac{I_{2st}}{I_2} = \left(\frac{s^2 + s_{mT}^2}{s^2\left(1 + s_{mT}^2\right)} \right)^{1/2}$	
	(b) $\frac{I_{2mT}}{I_2} = \left(\frac{1}{2}\left[1 + \left(\frac{s_{mT}}{s} \right)^2 \right] \right)^{1/2}$	
	OR	10
	Derive the maximum torque condition and prove that it is independent of rotor resistance. Also determine the slip at which maximum torque occurs and explain the physical significance of this condition.	6
8)	Writes the short notes on any two of the following a) Deep bar and double cage rotor b) Phasor groups of three-phase transformer. c) Cogging and Crawling	5+5